Neural Intelligence States: Exactly Mergeable Evidence Replaces Trained-Adapter Averaging for Compute-Efficient Distributed Adaptation

Draft — set author list before submission

Code and all measurements: github.com/tonyvone/rtlmai · July 2026 preprint draft

Abstract

Distributed adaptation of foundation models today means independently training low-rank adapters and averaging them—an approach that suffers from geometric misalignment between per-client subspaces, pays a full optimization run per model variant, and offers neither exact deletion nor forgetting-free incremental updates. We propose *neural intelligence states* (RTLM-N): instead of communicating trained weights, every node accumulates *sufficient-statistic evidence* for adaptation coordinates in a globally shared canonical basis anchored to a frozen backbone. States merge by addition—exactly, in any order—and models are *materialized* from the merged state by regularized linear solves. We show how to choose the shared basis (the optimal linear correction has rank at most the output dimension, spanned by a feature–residual cross-covariance computable on public calibration data), how to reach nonlinear internal surfaces via distributed iterated Gauss–Newton with per-round exact merging and measured linearization error, and how the same algebra yields three properties no trajectory-based method offers together: learning cost proportional to *new* data, zero forgetting, and exact machine unlearning by subtraction. On real text classification with genuinely Adam-trained competitors, canonical states score 0.750 versus 0.538 for federated LoRA under severe non-IID skew at 46× less measured compute; 1,000 model variants materialize from one state at 215× less measured compute than per-variant training; iterated Gauss–Newton reaches within 4.5 points of full backpropagation on an internal-only adaptation task from a 48-coordinate mergeable subspace; and a learning inference cascade cuts energy per task 2.6× within a single workload as teacher escalations decline. Every result carries an explicit guarantee class, and every number is reproducible from the released code.

1 Introduction

The economics of adapting foundation models are dominated by a structural choice: learning is stored in *trained weights*, produced by optimization trajectories over data. This choice taxes every downstream need. Producing M tenant- or domain-specialized variants costs M training runs. Keeping a deployed model current under drift costs either a replay over all accumulated history or an incremental fine-tune that forgets. Honoring a deletion request costs a retraining campaign or an approximation. Federating adaptation across data silos costs quality: independently trained low-rank adapters live in misaligned subspaces, and averaging them interferes destructively.

We revisit an older statistical idea—sufficient statistics—in the setting of frozen deep representations, and develop it into a complete distributed learning system. The central object is a *neural intelligence state*: a compact set of Gram and cross-moment accumulators, expressed in a canonical coordinate system shared by all nodes, from which models are *solved* rather than trained. Because the state is a sum, its algebra is exact: merging is associative, commutative, and order-free; deleting a contributor is a subtraction; and materialization depends only on the totals, never on how many raw examples produced them or in what order they arrived.

Contributions.

1. **Canonical evidence states.** We replace independently-optimized adapters with per-node sufficient statistics for coordinates in a globally fixed adaptation basis ($\Delta W = UCV^{\top}$ with U, V shared), and prove exact centralized equivalence for frozen-feature surfaces under arbitrary partitions, merge orders, and deletions (Prop. 1; measured deviation ~10^{-10} on real features).
2. **Principled basis selection.** We show the optimal linear correction has rank at most the output dimension and characterize its input subspace via a feature–residual cross-covariance (Prop. 2), yielding a data-aware shared basis computable from a small public calibration set. This choice is decisive in practice: random shared bases lose to learned-subspace LoRA, while cross-covariance bases match or beat it.
3. **Distributed iterated Gauss–Newton.** To adapt *internal* transformer surfaces beyond the one-shot linearization band, we iterate: re-linearize at the current coordinates, accumulate fresh Jacobian evidence, merge exactly within the round, and take a Levenberg–Marquardt damped step, reporting the measured linearization error every round. States from different linearization points are never merged—a claim-discipline rule enforced by construction.
4. **The incrementality trio.** The state algebra provides, simultaneously: learning cost O(new data) per refresh, zero forgetting (bit-identical to full retraining at every step), and exact unlearning by subtraction. We measure all three against replay retraining and warm-start SGD under distribution drift.
5. **A learning inference cascade.** Escalations to a large verified teacher are absorbed into the state as residual evidence, so the escalation rate is a declining curve rather than a routing constant; we measure declining teacher usage and energy per task at stable quality, and project fleet-scale energy with all assumptions declared.

Everything is released: a NumPy reference implementation, a PyTorch backbone with exact autograd Jacobians, sixteen machine-checked experiments, and per-claim guarantee labels (EXACT-FROZEN / LINEARIZED-BOUNDED / SKETCHED-BOUNDED / EMPIRICAL-NEURAL / UNSUPPORTED).

2 Related Work

Federated learning and adapter merging. FedAvg [13] averages locally trained weights; federated variants of LoRA [6] inherit the misalignment problem of averaging independently learned low-rank factors, as do post-hoc merging methods such as task arithmetic [7], model soups [17], and Fisher-weighted averaging [12]. Our approach differs in kind: nodes never train or transmit weights; they transmit evidence in a shared coordinate system, and the server solves. One-shot federated learning via distillation [3] shares the single-round goal but not the exactness.

Machine unlearning. Exact unlearning generally requires retraining or sharded architectures [2]; certified removal for linear models [4] is the closest antecedent to our deletion property. We extend exact removal to a fully distributed, mergeable, multi-surface setting with lineage and audit.

Continual learning. Regularization methods such as EWC [9] mitigate but do not eliminate forgetting. For supported surfaces our updates are exactly equivalent to retraining on the union of all data, so forgetting is structurally impossible rather than penalized.

Linearized and second-order methods. Our internal-surface machinery builds on Gauss–Newton and natural-gradient approximations [11] and on the linearization perspective of the NTK [8]; the trust-region damping is classical Levenberg–Marquardt [10]. The novelty is the distributed sufficient-statistic formulation with exact per-round merging and enforced per-round error reporting.

Cascades, distillation, and efficiency. Knowledge distillation [5] transfers teacher behavior once; inference cascades route between models. Our cascade *accumulates*: escalations become permanent state, addressing the energy concerns raised for large-scale ML [15, 14, 16]. Linear probes on frozen representations [1] underpin why the exact regime is practically useful and should become more so as backbones improve.

3 Method

3.1 Canonical evidence states

Let a frozen backbone produce representations $h = f_\theta(x) \in \mathbb{R}^d$, and let adaptation live in a globally shared basis: head updates $W = W_0 + UCV^\top$ with $U \in \mathbb{R}^{k\times r_u}$, $V \in \mathbb{R}^{d\times r_v}$ fixed and identical on every node, or generally $\Delta\theta = Pc$. For frozen-feature surfaces node i accumulates

$$H_i = \Sigma_j\, h_j h_j^\top, \quad R_i = \Sigma_j\, h_j r_j^\top,$$

where r_j is a target, teacher residual, or verified outcome; for low-rank surfaces the same moments in coordinates $z = V^\top h$, $e_U = U^\top(y - W_0 h)$. The global state is the sum over nodes, and materialization is a ridge solve, $W^* = (H + \lambda I)^{-1}R$. States carry a content hash, an HMAC signature, contributor lineage (for duplicate rejection), event commitments (for audited deletion), and a guarantee class; merging incompatible bases, backbones, or objectives is refused.

Proposition 1 (Exactness and invariance). *For a declared frozen representation and ridge objective, materialization from the merged state equals centralized training on the pooled raw data, for every partition of the data across nodes and every merge order; and removing a contributor's state yields exactly the centralized solution on the retained data.* Proof: H, R are sums of per-example terms, so any partition and order yields the same totals; the strictly convex objective has a unique minimizer depending only on those totals. Removal subtracts exactly the removed examples' terms. ∎

3.2 Choosing the shared basis

The choice of U, V decides whether the constrained space can express the needed correction.

Proposition 2 (Optimal correction subspace). *Let $\Delta W^* = \operatorname{argmin}_{\Delta W} \Sigma_j \|y_j - (W_0+\Delta W)h_j\|^2 + \lambda\|\Delta W\|^2_F$. Then $\Delta W^* = (H+\lambda I)^{-1}C_{xy}$ with $C_{xy} = \Sigma_j\, h_j e_j^\top$, $e_j = y_j - W_0 h_j$; in particular $\operatorname{rank}(\Delta W^*) \le k$ and its input subspace is spanned by the columns of $(H+\lambda I)^{-1}C_{xy}$.*

Thus a rank-k shared input basis suffices in principle. We estimate it from a small *public* calibration set: leading directions from the SVD of C_{xy} (a preconditioner-free surrogate), padded with top activation-variance directions and orthonormalized. Empirically this is the difference between losing and winning against learned-subspace LoRA (§4.2).

3.3 Internal surfaces: distributed iterated Gauss–Newton

For an internal surface (e.g., a transformer MLP weight, $\Delta W_1 = UCV^\top$), the network is linearized in the canonical coordinates: with $J(x) = \partial f/\partial c$ (computed exactly by autograd), nodes accumulate $G_i = \Sigma J^\top J$, $g_i = \Sigma J^\top r$, and the merged state yields $c^* = (G+\lambda I)^{-1} g$ (LINEARIZED-BOUNDED, with the linearization error measured on a probe set and attached to the artifact). One-shot solves are confined to a band around the reference model. We therefore iterate: at round t, every node re-linearizes at the current c_t, fresh evidence is accumulated and merged exactly *within the round*, and a damped step is taken with $\lambda_{eff} = \lambda + \mu \cdot \mathrm{tr}(G)/\dim(c)$ (Levenberg–Marquardt). Rounds are distinct objectives; states from different linearization points are never merged. Communication per round is one compact state per node.

3.4 Materialization, accretion, and migration

Factory. One merged state materializes arbitrarily many variants—regularization sweeps, tenant or geography slices via the state algebra, contributor exclusions, reduced-rank power-budget variants (exact sub-solves in leading canonical coordinates), and quantized models (guarantee demoted accordingly)—with no access to raw data and no representation passes. **Cascade.** A small edge model answers first; a verifier gates on confidence; failures escalate to a large verified teacher whose response is folded into the state as residual evidence; the correction rematerializes periodically. **Migration.** When the backbone changes, a linear representation bridge fit on public anchors transports heads ($W' = BW$) or states ($H' = B^\top HB$); transported objects are labeled EMPIRICAL-NEURAL, carry the measured bridge residual, and are never claimed exact.

4 Experiments

Setup and scope. We evaluate on AG News [18] (120k real news texts, 4 classes) with transformer backbones MLM-pretrained on the task corpus and then frozen (the build environment cannot reach pretrained-checkpoint hosts; the released code includes a HuggingFace wrapper exposing identical interfaces). All baselines that claim SGD are genuinely trained with Adam, including backpropagation through the backbone for internal LoRA. Compute is measured process CPU-seconds; energy is CPU-seconds at a declared wattage (RAPL where exposed). A NumPy reference program reproduces every algebraic claim independently of PyTorch. We report the guarantee class with every number.

4.1 Exactness on real representations

Across 10 nodes under severe label skew (Dirichlet α=0.1), arbitrary merge orders, node removal, and event deletion, merged-state materialization matches centralized ridge training to a maximum deviation of 2×10^{-10} on real MLM features (EXACT-FROZEN); state size is $O(d^2)$, independent of the number of examples (29× smaller than the raw data in the reference configuration, and constant as data grows).

4.2 Canonical states vs. federated adaptation

Eight non-IID nodes (α=0.1), identical frozen features for all methods, one-hot targets, accuracy on a held-out test set:

Method	*Accuracy*	*Measured cpu-s*
Base head (400 labels)	0.663	—
Federated LoRA (Adam, per-node subspaces, averaged)	0.538	4.67
One-shot FedAvg heads (Adam, averaged)	0.557	3.97
Centralized LoRA (Adam, all data)	0.742	1.78
Canonical evidence state (merged, solved)	**0.750**	**0.10**

The shared basis matters: with random shared directions the canonical state scores only 0.462–0.475; with the cross-covariance basis of §3.2 it beats even centralized SGD LoRA at 46× less measured compute, while remaining one-shot, mergeable, and deletable. In a controlled reference experiment where the true correction lies in the canonical space, averaging adapters trained in per-node misaligned bases scores 0.745 versus 0.990 for canonical merging—the geometric interference failure reproduced and then removed.

4.3 One state, one thousand models

From a single merged state we materialize 1,000 variants (regularization sweeps × contributor exclusions × quantization) in 7.8 measured cpu-s with zero representation passes, versus ~1,675 cpu-s for per-variant SGD training: **215×** (EXACT-FROZEN). Excluded-contributor variants are bit-identical to retraining without that contributor.

4.4 Internal adaptation: iterated Gauss–Newton vs. backprop

Internal MLP adapter only, head fixed, six non-IID nodes:

Method	*Accuracy*
Fixed head, no adapter (floor)	0.566
Federated internal LoRA (Adam, backprop, averaged)	0.648
One-shot Gauss–Newton state	0.666
Iterated Gauss–Newton (3 rounds)	**0.710**
Centralized internal LoRA (Adam, backprop)	0.755

Iteration escapes the one-shot band with per-round linearization errors 0.26 → 0.14 (measured, attached). A fixed 48-coordinate mergeable subspace closes to within 4.5 points of 1,536 freely-trained parameters, and beats the federated version of the real thing. Honest cost: Jacobian accumulation makes our method ~4× more expensive in cpu-s than SGD at this scale—the trade is compute for exact mergeability.

4.5 Incrementality: O(new), zero forgetting, exact deletion

Six periods of drifting data (70% class skew per period). The incremental state (i) stays bit-identical to full retraining at every period (max deviation $6{\times}10^{-11}$); (ii) cannot forget, while warm-start SGD on new data collapses to 0.482 vs 0.770; (iii) keeps per-period cost flat (×1.6 over six periods) while replay retraining grows ×6.0, for a cumulative measured advantage of 3.4× and diverging; and (iv) deletes a mid-stream contributor by subtraction at 269× less cost than replay-without-the-data, exactly. No trajectory-based method in the comparison offers more than two of these four properties at once.

4.6 Continual inference accretion

Feeding a repeated workload through the cascade, teacher escalation declines 0.37 → 0.13 at quality 1.000 (reference program; 0.36 → 0.30 with quality 0.62 → 0.86 on real text), and measured energy per accepted task falls 0.018 → 0.007 J, yielding 7.7× verified outcomes per joule versus teacher-always. A fleet-scale projection composing these measured micro-costs with a declared workload model (labeled simulation) indicates −81% lifetime energy versus teacher-always and −51% versus a tuned static cascade over 24 months, with backbone upgrades absorbed by state migration (below).

4.7 Backbone migration

Transporting the head state to an independently pretrained, narrower backbone through a bridge fit on 600 public anchors recovers 78% of the cold-reset→full-retrain gap (reset 0.229, transported 0.500, retrain 0.578; bridge residual 0.603 reported; EMPIRICAL-NEURAL).

5 Limitations

All results use small backbones pretrained on the task corpus; scale is untested (though the mechanism should benefit from stronger representations, which capture more of downstream adaptation linearly). Evidence is classification/scoring-shaped; generative tasks are untouched. Energy is measured CPU-time at declared wattage, not wall power; the fleet projection is a simulation and labeled as such. Migration is empirical. Byzantine-robust aggregation of validly signed but malicious statistics is future work. Exactness claims hold only for the declared frozen or linearized objectives—never for unrestricted end-to-end training.

6 Conclusion

Replacing trained-adapter communication with exactly mergeable evidence in a shared canonical basis turns distributed adaptation from an optimization problem into an algebra: models become cheap materialized views of a durable intelligence state that grows by addition, forgets nothing, deletes exactly, iterates to near-backprop quality on internal surfaces, and progressively transfers inference from large central models to small edge models. The measured savings are one to two orders of magnitude on the adaptation side and compounding on the serving side. The decisive open test—identical machinery on large pretrained backbones—requires only a representation extractor swap, which the released code already supports.

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